

Enumeration of colored permutations by the parity of descent positions

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Leonhard Euler (1707-1783)

Euler's triangle

$$\begin{array}{ccccccc} & & & & 1 & & \\ & & & 1 & & 1 & \\ & & 1 & & 4 & & 1 \\ & 1 & & 11 & & 11 & 1 \\ 1 & & 26 & & 66 & & 26 & 1 \end{array}$$

Euler's definition

$$\sum_{i \geq 1} i^n t^i = \frac{t \cdot A_n(t)}{(1-t)^{n+1}}$$

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$$\sum_{i \geq 1} i^n t^i = \frac{t \cdot A_n(t)}{(1-t)^{n+1}}$$

Euler's exponential generating function formula

$$\sum_{n \geq 0} A_n(t) \frac{z^n}{n!} = \frac{1-t}{e^{(t-1)z} - t}$$

A combinatorial interpretation of Eulerian numbers

$\mathfrak{S}_n$: Set of permutations of $[n] := \{1, \dots, n\}$.

For $\sigma \in \mathfrak{S}_n$,

Descent set:

$$\text{DES}(\sigma) := \{i \in [n-1] : \sigma_i > \sigma_{i+1}\}$$

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Define

$$\text{des}(\sigma) := |\text{DES}(\sigma)|$$

$$\text{asc}(\sigma) := |\text{ASC}(\sigma)|.$$

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Define

$$\text{des}(\sigma) := |\text{DES}(\sigma)|$$

$$\text{asc}(\sigma) := |\text{ASC}(\sigma)|.$$



Major Percy Alexander
MacMahon (1854-1929)

Example

$$\text{des}(3715624) = |\{2, 5\}| = 2$$

$$\text{asc}(3715624) = |\{1, 3, 4, 6\}| = 4$$



P. A. MacMahon, *Combinatory analysis*,

vols 1 and 2, Cambridge University Press, 1916.

A combinatorial interpretation of Eulerian numbers

$\mathfrak{S}_3$	des	asc
123	0	2
132	1	1
213	1	1
231	1	1
312	1	1
321	2	0

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$$\sum_{\sigma \in \mathfrak{S}_3} t^{\text{des}(\sigma)} = 1 + 4t + t^2$$

$$\sum_{\sigma \in \mathfrak{S}_3} t^{\text{asc}(\sigma)} = 1 + 4t + t^2$$

Euler's triangle

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$\mathfrak{S}_3$	des	asc	
123	0	2	$\sum_{\sigma \in \mathfrak{S}_3} t^{\text{des}(\sigma)} = 1 + 4t + t^2$
132	1	1	
213	1	1	
231	1	1	$\sum_{\sigma \in \mathfrak{S}_3} t^{\text{asc}(\sigma)} = 1 + 4t + t^2$
312	1	1	
321	2	0	

Euler's triangle

1
1 1
1 4 1
1 11 11 1
1 26 66 26 1

Theorem (Riordan 1958)

$$A_n(t) = \sum_{j=0}^{n-1} \left\langle \begin{matrix} n \\ j \end{matrix} \right\rangle t^j = \sum_{\sigma \in \mathfrak{S}_n} t^{\text{des}(\sigma)}.$$

where $\left\langle \begin{matrix} n \\ j \end{matrix} \right\rangle := |\{\sigma \in \mathfrak{S}_n : \text{des}(\sigma) = j\}|$.

Inversion and q -analogue

Let $\sigma \in \mathfrak{S}_n$,

Inversion number:

$$\text{inv}(\sigma) := |\{(i, j) : \sigma(i) > \sigma(j), 1 \leq i < j \leq n\}|.$$

$$\text{inv}(3142) = 3$$

Define q -analogue of n and n factorial:

$$[n]_q := 1 + q + q^2 + \cdots + q^{n-1}$$

$$[n]_q! := [n]_q [n-1]_q \cdots [1]_q$$

$\mathfrak{S}_3$	inv
123	0
132	1
213	1
231	2
312	2
321	3

$$\begin{aligned} & \sum_{\sigma \in \mathfrak{S}_3} q^{\text{inv}(\sigma)} \\ &= 1 + 2q + 2q^2 + q^3 \\ &= (1 + q + q^2)(1 + q) \end{aligned}$$

Theorem (Rodrigues 1839)

$$\sum_{\sigma \in \mathfrak{S}_n} q^{\text{inv}(\sigma)} = [n]_q!$$

q -analogue of Euler's exp. generating function formula

Theorem (Stanley 1976)

We have

$$\sum_{n \geq 0} \frac{t^n}{[n]_q!} \sum_{\sigma \in \mathfrak{S}_n} x^{\text{des}(\sigma)} q^{\text{inv}(\sigma)} = \frac{(1-t) \exp(t(1-x); q)}{1 - t \exp(t(1-x); q)},$$

where

$$\exp(t; q) = \sum_{n \geq 0} \frac{t^n}{[n]_q!}.$$

When $q = 1$, it reduces to Euler's formula.



R. P. Stanley, Binomial posets, Möbius inversion, and permutation enumeration,

J. Combin. Thy. A, **20** (1976), 336–356.

Variation on the descent statistic

Let $\sigma \in \mathfrak{S}_n$,

Odd descent set:

$$\text{ODES}(\sigma) := \{i \in [n-1] : \sigma_i > \sigma_{i+1} \text{ and } i \text{ is odd}\}$$

$$\sigma = 32541, \text{ ODES}(\sigma) = \{1, 3\}$$

Even descent set:

$$\text{EDES}(\sigma) := \{i \in [n-1] : \sigma_i > \sigma_{i+1} \text{ and } i \text{ is even}\}$$

$$\sigma = 32541, \text{ EDES}(\sigma) = \{4\}$$

Alternating descent set:

$$\text{ALDES}(\sigma) := \{i \in [n-1] : \sigma_i > \sigma_{i+1} \text{ and } i \text{ is odd or } \sigma_i < \sigma_{i+1} \text{ and } i \text{ is even}\}.$$

$$\sigma = 32541, \text{ ALDES}(\sigma) = \{1, 2, 3\}$$

Chebikin introduced in 2008.

Refinement of Eulerian polynomials

In 1973, Carlitz and Scoville considered the **four-variable** polynomials

$$P_n(x_0, x_1, y_0, y_1) = \sum_{\sigma \in \mathfrak{S}_n} x_0^{\text{easc}(\sigma)} x_1^{\text{oasc}(\sigma)} y_0^{\text{edes}(\sigma)} y_1^{\text{odes}(\sigma)}.$$

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Theorem (Carlitz and Scoville 1973)

Let $\alpha = \sqrt{(y_0 - x_0)(y_1 - x_1)}$, we have

$$\sum_{n \geq 1} P_n(x_0, x_1, y_0, y_1) \frac{t^n}{n!} = \frac{(x_1 + y_1) \sum_{n \geq 1} \frac{\alpha^{n-1} t^{2n}}{(2n)!} + \sum_{n \geq 1} \frac{\alpha^{n-1} t^{2n-1}}{(2n-1)!}}{1 - (x_0 y_1 + x_1 y_0) \sum_{n \geq 1} \frac{\alpha^{n-1} t^{2n}}{(2n)!}}.$$

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In 2008, Chebikin considered the **alternating** Eulerian polynomials

$$\sum_{\sigma \in \mathfrak{S}_n} t^{\text{aldes}(\sigma)}.$$

In 2018, Sun considered the **bivariate** Eulerian polynomials

$$\sum_{\sigma \in \mathfrak{S}_n} p^{\text{odes}(\sigma)} q^{\text{edes}(\sigma)}.$$

q -analogue of Carlitz and Scoville's formula

Let

$$P_n(x_0, x_1, y_0, y_1, q) := \sum_{\sigma \in \mathfrak{S}_n} x_0^{\text{easc}(\sigma)} x_1^{\text{oasc}(\sigma)} y_0^{\text{edes}(\sigma)} y_1^{\text{odes}(\sigma)} q^{\text{inv}(\sigma)}.$$

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Recall that the q -trigonometric series is defined by

$$\cosh(t; q) = \sum_{n \geq 0} \frac{t^{2n}}{[2n]_q!}, \quad \sinh(t; q) = \sum_{n \geq 0} \frac{t^{2n+1}}{[2n+1]_q!}.$$

Theorem (Pan and Zeng 2022)

Let $\alpha = \sqrt{(y_0 - x_0)(y_1 - x_1)}$, we have

$$\begin{aligned} & \sum_{n \geq 1} P_n(x_0, x_1, y_0, y_1, q) \frac{t^n}{[n]_q!} \\ &= \frac{(x_1 + y_1) \cosh(\alpha t; q) + \alpha \sinh(\alpha t; q) - y_1(\cosh^2(\alpha t; q) - \sinh^2(\alpha t; q)) - x_1}{x_0 x_1 - (x_0 y_1 + x_1 y_0) \cosh(\alpha t; q) + y_0 y_1 (\cosh^2(\alpha t; q) - \sinh^2(\alpha t; q))}. \end{aligned}$$

When $q = 1$, it reduces to Carlitz and Scoville's formula.

Coxeter group

Define a **Coxeter system** to be a pair (W, S) , where W is a group, and S is a minimal generating set of W .

For every s, t in S , satisfying the following relations:

- $st = e$ if and only if $s = t$, and if $s \neq t$;
- $(st)^{m(s,t)} = (ts)^{m(s,t)} = e$ for some integer $m(s, t) > 1$.

We say such a group W is a **Coxeter group** of rank $r = |S|$.

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For $n \geq 2$, the symmetric group $\mathfrak{S}_n$ has a Coxeter presentation with generators $\{\tau_1, \dots, \tau_{n-1}\}$ and defining relations

- $\tau_i^2 = e$ for $1 \leq i \leq n - 1$;
- $\tau_i \tau_{i+1} \tau_i = \tau_{i+1} \tau_i \tau_{i+1}$ for $1 \leq i \leq n - 2$;
- $\tau_i \tau_j = \tau_j \tau_i$ for $|i - j| \geq 2$.

where τ_i is meaning that **swapping** the entries in positions i and $i + 1$.

Coxeter group of type A , what's more, type $B, D, E \dots$

Analogy with symmetric group (length, descent)

For $w \in W$, define

Length:

$$\ell(w) = \min\{k \in \mathbb{N} : w = s_1 \cdots s_k, s_i \in S\}.$$

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Descent set:

$$\text{DES}(w) = \{s \in S : \ell(w) > \ell(ws)\}.$$

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When $W = \mathfrak{S}_n$,

Proposition (Bjorner and Brenti 2005)

$$\text{inv}(w) = \ell(w).$$

Proposition (Bjorner and Brenti 2005)

$$\text{DES}(w) = \{s_i \in \mathfrak{S}_n : w(i) > w(i+1)\}.$$



A. Björner and F. Brenti, *Combinatorics of Coxeter groups*,
G.T.M 231, Springer-Verlag, New York, 2005.

(B_n, S_B) Coxeter group of type B

Let B_n be the set of all bijections σ of the set $[\pm n]$ in itself such that

$$\sigma(-i) = -\sigma(i)$$

for all $i \in [\pm n]$, with the composition as group operation.

Window notation: $\sigma = [a_1, \dots, a_n]$, such that $\sigma(i) = a_i$.

The group is often called the group of all "**signed permutations**" of $[n]$.

Generator set: $S_B := \{s_1^B, \dots, s_{n-1}^B, s_0^B\}$, where

$$s_i^B = [1, \dots, i-1, i+1, i, i+2, \dots, n]$$

for $i \in [n-1]$, and

$$s_0^B = [-1, 2, \dots, n]$$

Length: $\ell_B(\sigma) = \min\{r \in \mathbb{N} \mid \sigma = s_{i_1}^B \cdots s_{i_r}^B, s_{i_j}^B \in S_B\}$.

B -descent set: $\text{DES}_B(\sigma) := \{s \in S_B : \ell_B(\sigma) > \ell_B(\sigma s)\}$.

Type B analogue of Carlitz and Scoville's formula

Combinatorial definition:

Length: $\ell_B(\sigma) = \text{inv}(\sigma) - \sum_{\sigma(i) < 0} \sigma(i)$.

B -descent set: $\text{DES}_B(\sigma) = \{i \in [n-1] : \sigma(i-1) > \sigma(i)\}$.

Pan and Zeng considered

$$B_n(x, y) := \sum_{\sigma \in \mathcal{B}_n} x^{\text{odes}_B(\sigma)} y^{\text{edes}_B(\sigma)}.$$

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In a follow up, Dey et al. gave a q -analogue version

$$B_n(x, y, q) := \sum_{\sigma \in \mathcal{B}_n} x^{\text{odes}(\sigma)} y^{\text{edes}(\sigma)} q^{\ell_B(\sigma)}.$$

Note that Coxeter group of type A and B are special cases of Wreath product $\mathbb{Z}_r \wr \mathfrak{S}_n$.

Our results will unify the formulas of type A and B .

Wreath product and r -colored permutations

Wreath product $\mathbb{Z}_r \wr \mathfrak{S}_n$

$$G(r, n) := \{(c_1, \dots, c_n; \sigma) \mid c_i \in [0, r-1], \sigma \in \mathfrak{S}_n\}.$$

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Multiplication:

$$(c; \sigma) \cdot (c'; \tau) := ((c_1 + c'_{\tau^{-1}(1)}, \dots, c_n + c'_{\tau^{-1}(n)}); \sigma \circ \tau),$$

where addition $+$ is in $\mathbb{Z}_r$ and composition $\circ$ is in $\mathfrak{S}_n$.

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Window notation:

$$\gamma = [\sigma(1)^{c_1}, \dots, \sigma(n)^{c_n}] \xrightarrow{\text{simplification}} [\gamma(1), \dots, \gamma(n)]$$

The entry c_i is called the color of the $\sigma(i)$.

Viewed as a r -colored permutation, introduced by Steingrímsson.

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Example

$$\gamma = [3^1, 2, 1^3, 4^2, 6^2, 5^1] \in G(5, 6).$$



E. Steingrímsson. Permutation statistics of indexed permutations,
Eur. J. Combin. **15**, (1994), 187–205.

Generator set, length and descent set

Generator set: $S_G := \{s_0, s_1, \dots, s_{n-1}\}$, where for $i \in [n-1]$

$$s_i := [1, \dots, i-1, i+1, i, i+2, \dots, n] \text{ and } s_0 := [1^1, 2, \dots, n].$$

For $\gamma \in G(r, n)$,

Length:

$$\ell_G(\gamma) := \min\{r \in \mathbb{N} : \gamma = s_{i_1} \cdots s_{i_r}, \text{ for some } s_{i_j} \in S_G\}.$$

Descent set:

$$\text{DES}_G(\gamma) := \{s \in S_G : \ell_G(\gamma s) < \ell_G(\gamma)\}$$

and its size will be denoted by $\text{des}_G(\gamma)$.

Combinatorial definition of length

Linear order:

$$n^{r-1} < \dots < n^1 < \dots < 1^{r-1} < \dots < 1^1 < 0 < 1 < \dots < n.$$

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For $\gamma \in G(r, n)$,

Length: (Combinatorial definition)

$$\ell_G(\gamma) := \text{inv}(\gamma) + \sum_{c_i \neq 0} (\sigma(i) + c_i - 1),$$

where $\text{inv}(\gamma) = |\{(i, j) : 1 \leq i < j \leq n, \text{ and } \gamma(i) > \gamma(j)\}|$.

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When $r = 1$, it reduces to $\text{inv}(\gamma)$ in $\mathfrak{S}_n$.

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Length: (Combinatorial definition)

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where $\text{inv}(\gamma) = |\{(i, j) : 1 \leq i < j \leq n, \text{ and } \gamma(i) > \gamma(j)\}|$.

When $r = 1$, it reduces to $\text{inv}(\gamma)$ in $\mathfrak{S}_n$.

When $r = 2$, it reduces to the length in type B Coxeter group

$$\ell_B(\gamma) := \text{inv}(\gamma) + \sum_{c_i \neq 0} \sigma(i).$$

Combinatorial definition of descent, colored set

For $\gamma \in G(r, n)$,

Descent set:

$$\text{DES}_G(\gamma) := \{i \in [0, n-1] : \gamma(i) > \gamma(i+1)\},$$

where $\gamma(0) := 0$. Note that $0 \in \text{DES}_G(\gamma)$ iff $c_1 > 0$.

Combinatorial definition of descent, colored set

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where $\gamma(0) := 0$. Note that $0 \in \text{DES}_G(\gamma)$ iff $c_1 > 0$.

Color set:

$$\text{COL}(\gamma) := \{i \in [n] : c_i > 0\}$$

Define $\text{des}_G(\gamma) = |\text{DES}_G(\gamma)|$ and $\text{col}(\gamma) = |\text{COL}(\gamma)|$

Example

If $\gamma = [3^1, 2, 1^3, 4^2, 6^2, 5^1] \in G(5, 6)$, then $\text{DES}_G(\gamma) = \{0, 2, 3, 4\}$, $\text{des}_G(\gamma) = 4$, $\ell_G(\gamma) = 34$ and $\text{col}(\gamma) = 5$.

A q -analogue of exponential series over wreath product

For $n \in \mathbb{N}$,

$$(a; p)_n := \begin{cases} 1, & \text{if } n = 0; \\ (1 - a)(1 - ap) \cdots (1 - ap^{n-1}), & \text{if } n \geq 1. \end{cases}$$

Theorem

$$G_{(r,n)}(a, q) := \sum_{\gamma \in G(r,n)} a^{\text{col}(\gamma)} q^{\ell_G(\gamma)} = [n]_q! (-aq[r-1]_q; q)_n.$$

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$$(a; p)_n := \begin{cases} 1, & \text{if } n = 0; \\ (1 - a)(1 - ap) \cdots (1 - ap^{n-1}), & \text{if } n \geq 1. \end{cases}$$

Theorem

$$G_{(r,n)}(a, q) := \sum_{\gamma \in G(r,n)} a^{\text{col}(\gamma)} q^{\ell_G(\gamma)} = [n]_q! (-aq[r-1]_q; q)_n.$$

q -analogue of exponential series over wreath product, that is

$$\exp_{G(r)}(t; a, q) := \sum_{n \geq 0} \frac{t^n}{G_{(r,n)}(a, q)}$$

and write

$$\cosh_{G(r)}(t; a, q) = \frac{\exp_{G(r)}(t; a, q) + \exp_{G(r)}(-t; a, q)}{2},$$

$$\sinh_{G(r)}(t; a, q) = \frac{\exp_{G(r)}(t; a, q) - \exp_{G(r)}(-t; a, q)}{2}.$$

Define the enumerative polynomials

$$G_{(r,n)}(x, y, a, q) := \sum_{\gamma \in G(r,n)} x^{\text{odes}_G(\gamma)} y^{\text{edes}_G(\gamma)} a^{\text{col}(\gamma)} q^{\ell_G(\gamma)},$$

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When $r = 1$, it reduces to type A. When $r = 2$, it reduces to type B. Define their generating functions

$$G_0 := \sum_{n \geq 0} G_{(r,2n)}(x, y, a, q) \frac{t^{2n}}{G_{(r,2n)}(a, q)},$$

$$G_1 := \sum_{n \geq 0} G_{(r,2n+1)}(x, y, a, q) \frac{t^{2n+1}}{G_{(r,2n+1)}(a, q)}.$$

Theorem

For any $n \in \mathbb{N}$, we have

$$G_0 = (1 - y) \cdot$$

$$\frac{\left((1 - x \cosh(Mt; q)) \cosh_{G(r)}(Mt; a, q) + x \sinh(Mt; q) \sinh_{G(r)}(Mt; a, q) \right)}{1 - (x + y) \cosh(Mt; q) + xy \exp(Mt; q) \exp(-Mt; q)},$$

$$G_1 = M \cdot$$

$$\frac{\left((1 - y \cosh(Mt; q)) \sinh_{G(r)}(Mt; a, q) + y \sinh(Mt; q) \cosh_{G(r)}(Mt; a, q) \right)}{1 - (x + y) \cosh(Mt; q) + xy \exp(Mt; q) \exp(-Mt; q)}.$$

$$\text{where } M = \sqrt{(1 - x)(1 - y)}.$$

- When $r = 1$, $a = 0$, we can obtain the formula of Pan and Zeng about type A .
- When $r = 2$, $a = 1$, we can obtain the formula of Dey et al. about type B .
- When $r = 2$, $x = y$, $q = 1$, then substituting $t \leftarrow 2t$, we obtain Brenti's formula,

$$\sum_{n \geq 0} G_{(2,n)}(x, x, a, 1) \frac{t^n}{n!} = \frac{(1-x) \exp(t(1-x))}{1-x \exp(t(1-x)(1+a))}.$$

Some notations

$\binom{[n]}{m}^r$: the set of all r -colored subsets of size m of $[n]$.

For $A^r = \{a_1^{c_1}, a_2^{c_2}, \dots, a_m^{c_m}\} < \in \binom{[n]}{m}^r$,

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$G_i(r, n)$: set of colored permutations in $G(r, n)$ with the last $n - i$ elements ordered increasingly

$$G_i(r, n) := \{\gamma \in G(r, n) : \gamma(i+1) < \dots < \gamma(n-1) < \gamma(n)\}.$$

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If $\gamma = [\sigma(1)^{c_1}, \sigma(2)^{c_2}, \dots, \sigma(n-m)^{c_{n-m}}] \in G(r, n-m)$ and $A^r \in \binom{[n]}{m}^r$ with $A \subseteq [n]$ and $[n] \setminus A = \{s_1, s_2, \dots, s_{n-m}\}_{<}$, define a mapping $f : G(r, n-m) \times \binom{[n]}{m}^r \rightarrow G_{n-m}(r, n)$ by

$$f(\gamma, A^r) = \gamma|_{[n] \setminus A} * [A^r].$$

where $[u] * [v] := [uv]$ and

$$\gamma|_{[n] \setminus A} := [s_{\sigma(1)}^{c_1}, s_{\sigma(2)}^{c_2}, \dots, s_{\sigma(n-m)}^{c_{n-m}}].$$

Alternative definition of length

For $\gamma = [\gamma(1), \dots, \gamma(n)] = [\sigma(1)^{c_1}, \dots, \sigma(n)^{c_n}] \in G(r, n)$, Define

$$\text{inv}_c(\gamma) = \sum_{1 \leq i < j \leq n} |\{(i, j) : \gamma_c(i) > \gamma_c(j)\}|,$$

where $\gamma_c(i)$ is defined by

$$\gamma_c(i) = \begin{cases} |\gamma(i)| & \text{if } c_i \neq 0; \\ |\gamma(i)|^1 & \text{if } c_i = 0. \end{cases}$$

Lemma

For $\gamma = [\gamma(1), \dots, \gamma(n)] = [\sigma(1)^{c_1}, \dots, \sigma(n)^{c_n}] \in G(r, n)$, we have

$$\ell_G(\gamma) = \text{inv}(\gamma) + \text{inv}_c(\gamma) + C(\gamma),$$

with $C(\gamma) = \sum_{c_i \neq 0} c_i$.

mapping f satisfies

Lemma

For $0 \leq m \leq n$, let $\gamma \in G(r, n-m)$ and $A^r \in \binom{[n]}{m}^r$. Then the mapping $f : G(r, n-m) \times \binom{[n]}{m}^r \rightarrow G_{n-m}(r, n)$ satisfies

$$\begin{aligned}C(f(\gamma, A^r)) &= C(\gamma) + C(f(e, A^r)), \\ \text{col}(f(\gamma, A^r)) &= \text{col}(\gamma) + \text{col}(f(e, A^r)), \\ \ell_G(f(\gamma, A^r)) &= \ell_G(\gamma) + \ell_G(f(e, A^r)),\end{aligned}$$

where e is the identity permutation in $G(r, n-m)$.

Proof.

The first two identities are obvious. The second property can be proved by an alternate definition of length. □

$$\text{col}(f(\gamma, A^r), \ell_G(f(\gamma, A^r)))$$

Lemma

Let $0 \leq m \leq n$. For any $\gamma \in G(r, n - m)$, we have

$$\sum_{A^r \in \binom{[n]}{m}^r} a^{\text{col}(f(\gamma, A^r))} q^{\ell_G(f(\gamma, A^r))} = a^{\text{col}(\gamma)} q^{\ell_G(\gamma)} \binom{n}{m}_q (-aq^{n-m+1}[r-1]_q; q)_m.$$

Proof.

It suffices to prove the identity for $\gamma := e$. Let

$$F_m(a, q) := \sum_{A^r \in \binom{[n]}{m}^r} a^{\text{col}(f(e, A^r))} q^{\ell_G(f(e, A^r))}.$$

Then we have to prove the following identity

$$[n - m]_q! (-aq[r - 1]_q; q)_{n-m} F_m(a, q) [m]_q! = [n]_q! (-aq[r - 1]_q; q)_n.$$



Sketch of the Proof

Proof.

For $0 < m \leq n$, if $\gamma = [\sigma(1)^{c_1}, \sigma(2)^{c_2}, \dots, \sigma(n-m)^{c_{n-m}}] \in G(r, n-m)$, $A^r = \{a_1^{c'_1}, a_2^{c'_2}, \dots, a_m^{c'_m}\}_{<} \in \binom{[n]}{m}^r$, $[n] \setminus A = \{s_1, s_2, \dots, s_{n-m}\}_{<}$ and $\sigma = \sigma_1 \sigma_2 \cdots \sigma_m \in \mathfrak{S}_m$, we define a mapping

$\Phi : G(r, n-m) \times \binom{[n]}{m}^r \times \mathfrak{S}_m \rightarrow G(r, n)$ by

$$\Phi(\gamma, A^r, \sigma) = \gamma|_{[n] \setminus A} * \sigma(A^r),$$

where

$\sigma(A^r) := [a_{\sigma_1}^{c'_1}, a_{\sigma_2}^{c'_2}, \dots, a_{\sigma_m}^{c'_m}]$, and $\gamma|_{[n] \setminus A} := [s_{\sigma(1)}^{c_1}, s_{\sigma(2)}^{c_2}, \dots, s_{\sigma(n-m)}^{c_{n-m}}]$,

Φ is a bijection and satisfies

$$\begin{aligned} \text{col}(f(e, A^r)) + \text{col}(\gamma) &= \text{col}(\Phi(\gamma, A^r, \sigma)); \\ \ell_G(f(e, A^r)) + \ell_G(\gamma) + \text{inv}(\sigma) &= \ell_G(\Phi(\gamma, A^r, \sigma)). \end{aligned}$$



$G_i(r, n)$: the set of permutations with the last $n - i$ elements increasing.

Weight: $w(\gamma) = x^{\text{odes}_G(\gamma)} y^{\text{edes}_G(\gamma)} a^{\text{col}(\gamma)} q^{\ell_G(\gamma)}$.

Lemma

Let $n \in \mathbb{N}$ and $r \in \mathbb{P}$. The following identities hold.

① If i is odd,

$$\sum_{\gamma \in G_i(r, n)} w(\gamma) = x \frac{G_{(r, i)}(x, y, a, q) G_{(r, n)}(a, q)}{G_{(r, i)}(a, q) [n - i]_q!} - (x - 1) \sum_{\gamma \in G_{i-1}(r, n)} w(\gamma).$$

② If i is even,

$$\sum_{\gamma \in G_i(r, n)} w(\gamma) = y \frac{G_{(r, i)}(x, y, a, q) G_{(r, n)}(a, q)}{G_{(r, i)}(a, q) [n - i]_q!} - (y - 1) \sum_{\gamma \in G_{i-1}(r, n)} w(\gamma).$$

Permutations of type D_n

Denote by $\mathcal{D}_n$ the subgroup of $\mathcal{B}_n$ (i.e. $G(2, n)$) consisting of all the colored permutations having an **even** number of colored entries, that is,

$$\mathcal{D}_n := \{\gamma \in \mathcal{B}_n : \text{col}(\sigma) \equiv 0 \pmod{2}\}.$$

We denote -1 by $\bar{1}$ and define $\gamma(\bar{1}) := -\gamma(1)$.

For $\gamma = [\gamma(1), \dots, \gamma(n)] \in \mathcal{D}_n$, define the **type D descent set** to be

$$\text{DES}_D(\gamma) := \{i \in \{-1, 1, 2, \dots, n\} : \gamma(i) > \gamma(|i| + 1)\},$$

and its cardinality is denoted by $\text{des}_D(\sigma)$.

The **length** in type D Coxeter group is defined as follows:

$$\ell_D(\gamma) := \ell_B(\gamma) - \text{col}(\gamma),$$

Consider the parity of descent position

Define the enumerative polynomials

$$D_n(x, y, a, q) := \sum_{\gamma \in \mathcal{D}_n} x^{\text{odes}_D(\gamma)} y^{\text{edes}_D(\gamma)} a^{\text{col}(\gamma)} q^{\ell_D(\gamma)},$$

The following theorem is our the second result.

Theorem

For any $n \in \mathbb{P}$, We have

$$\begin{aligned} D_{2n}(x, y, a, q) &= \frac{1}{2} \left(B_n(-a, q) \langle t^n \rangle \widehat{B}_0(t; x, y, -\frac{a}{q}, q) + B_n(a, q) \langle t^n \rangle \widehat{B}_0(t; x, y, \frac{a}{q}, q) \right), \\ D_{2n+1}(x, y, a, q) &= \frac{1}{2} \left(B_n(-a, q) \langle t^n \rangle \widehat{B}_1(t; x, y, -\frac{a}{q}, q) + B_n(a, q) \langle t^n \rangle \widehat{B}_1(t; x, y, \frac{a}{q}, q) \right), \end{aligned}$$

where the notation $\langle t^n \rangle f(t)$ stands for the coefficient of t^n in $f(t)$.

Signed Eulerian identity

Loday conjectured, Désarménien and Foata first investigated the relationship between a signed Eulerian polynomial and the classic Eulerian polynomial, called [signed Eulerian identity](#),

$$\sum_{\pi \in \mathfrak{S}_{2n}} (-1)^{\ell(\pi)} t^{\text{des}(\pi)} = (1-t)^n \sum_{\pi \in \mathfrak{S}_n} t^{\text{des}(\pi)}$$

and

$$\sum_{\pi \in \mathfrak{S}_{2n+1}} (-1)^{\ell(\pi)} t^{\text{des}(\pi)} = (1-t)^n \sum_{\pi \in \mathfrak{S}_{n+1}} t^{\text{des}(\pi)}.$$

Wachs proposed an elegant involution proof.



[J.L. Loday, Opérations sur l'homologie cyclique des algèbres commutatives.](#)

Invent. Math. **96** (1989) 205-230.



[J. Désarménien, D. Foata, The signed Eulerian numbers.](#)

Discrete Math. **99** (1992), no. 1-3, 49-58.



[M. Wachs, An involution for signed Eulerian numbers.](#)

Discrete Math. **99** (1992), no. 1-3, 59-62.

Counting signed alternating descents

Dey and Sivasubramanian computed the signed alternating descent enumeration in classical Weyl groups of types A and B.



H. K. Dey, and S. Sivasubramanian, Signed alternating descent enumeration in classical Weyl groups,

Discrete Math. **346** (2023), no. 10, Paper No. 113540.

We now evaluate the signed alternating descent enumeration for colored permutations.

Define the *signed alternating descent polynomials*

$$\text{Alt}_n^{G(r)}(x) := \sum_{\gamma \in G(r,n)} (-1)^{\ell_G(\gamma)} x^{\widehat{d}_G(\gamma)},$$

and q -alternating descent polynomials over wreath product by

$$\text{Alt}_n^{G(r)}(x, q) := \sum_{\gamma \in G(r,n)} x^{\widehat{d}_G(\gamma)} q^{\ell_G(\gamma)},$$

Counting signed alternating descent

Theorem

If $r \geq 2$ is an even integer, then

$$\text{Alt}_n^{G(r)}(x) = (-1)^{\lfloor (n+1)/2 \rfloor} (1-x)^n \quad \text{for } n \geq 0.$$

Theorem

If $r \geq 1$ is an odd integer, then $\text{Alt}_1^{G(r)}(x) = x$ and

$$\text{Alt}_n^{G(r)}(x) = \begin{cases} x(1-x)^m A_m(x) & \text{if } n = 2m \ (m \in \mathbb{N}^*); \\ \frac{2x^2}{1+x} (1-x)^{2m} A_{2m}(x) & \text{if } n = 4m + 1 \ (m \in \mathbb{N}^*); \\ 0 & \text{if } n = 4m + 3 \ (m \in \mathbb{N}). \end{cases}$$

When $r = 1, 2$, the above formulas reduce to Dey and Sivasubramanian's results.

Generating function for alternating descent

Note that $\text{Alt}_n^{G(r)}(x, q) := x^{\lfloor (n+1)/2 \rfloor} G_{(r,n)}(x, 1/x, 1, q)$.

Lemma

Let $\text{Alt}_0^{G(r)}(x; q) = 1$ and $M = 1 - x$. For positive integer r , we have

$$\sum_{n \geq 0} \frac{\text{Alt}_{2n}^{G(r)}(x, q)}{(-q[r-1]_q; q)_{2n}} \cdot \frac{t^{2n}}{[2n]_q!} = (x-1) \times \frac{(1 - x \cos(Mt; q)) \cos_{G(r)}(Mt; q) - x \sin(Mt; q) \sin_{G(r)}(Mt; q)}{x - (x^2 + 1) \cos(Mt; q) + x \exp(iMt; q) \exp(-iMt; q)},$$

$$\sum_{n \geq 0} \frac{\text{Alt}_{2n+1}^{G(r)}(x, q)}{(-q[r-1]_q; q)_{2n+1}} \cdot \frac{t^{2n+1}}{[2n+1]_q!} = (x-1) \times \frac{(x - \cos(Mt; q)) \sin_{G(r)}(Mt; q) + \sin(Mt; q) \cos_{G(r)}(Mt; q)}{x - (x^2 + 1) \cos(Mt; q) + x \exp(iMt; q) \exp(-iMt; q)}.$$

Sketch of the Proof

Multiplying the two sides by

$$x - (x^2 + 1) \cos((1-x)t; q) + x \exp(i(1-x)t; q) \cdot \exp(-i(1-x)t; q),$$

then taking the limit $q \rightarrow -1$, we verify that if $r \geq 2$ is an even integer, then

$$\text{Alt}_n^{G(r)}(x, -1) = (-1)^{\lfloor (n+1)/2 \rfloor} (1-x)^n \quad \text{for } n \geq 0.$$

Similarly for $r \geq 1$ is an odd integer, with $n = 2m, 4m + 1$. For $n = 4m + 3$, we construct a weight preserving and sign reversing involution to prove that $\text{Alt}_{4m+3}^{G(r)}(x, -1) = 0$.

Thanks